

A spectral collocation method for nonlocal diffusion equations with volume constrained boundary conditions



Hao Tian^{a,1}, Jing Zhang^{b,2,*}, Lili Ju^{c,a}

^a School of Mathematical Sciences, Ocean University of China, Qingdao, Shandong 266100, China

^b School of Mathematics and Statistics & Hubei Key Laboratory of Mathematical Sciences, Central China Normal University, Wuhan 430079, China

^c Department of Mathematics, University of South Carolina, Columbia, SC 29208, USA

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ABSTRACT

The nonlocal diffusion model describes the diffusion process of solutes in complex media properly, while the classical theory of partial differential equations can not provide an appropriate description. The purpose of this paper is to provide illustrations from both theoretical and numerical perspectives of the computation of nonlocal diffusion models by Legendre collocation methods. Compared to local numerical methods, Legendre collocation methods can achieve a fixed accuracy with much fewer unknowns whenever the computational domain is regular and the solutions are sufficiently smooth. This paper is a groundwork towards efficient high order methods including spectral and spectral element methods for nonlocal diffusion equations with volume constrained boundary conditions.

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1. Introduction

Nonlocal models have gained significant popularity in many fields for decades, such as physics, material science, social science [18,23,30–35], phase transitions [2,25], nonlocal heat conduction [3], convection-diffusion processes [6,39,40], kinetic equations [21], and image analysis [5,16,22]. These models allow their solutions to develop spatial singularities that can be used to characterize the effects of various types of heterogeneities and defects. An illustrative example is the nonlocal diffusion model [8], which can be applied to describe the statistical nature of a stochastic process [10].

Let $\mathbb{R}^d$ be the d -dimensional Euclidean space. For $\mathbf{x} \in \mathbb{R}^d$, we write $\mathbf{x} = (x_1, \dots, x_d)^t$ as a column vector. Define $\mathcal{L}_\delta$ as a nonlocal diffusion operator that represents nonlocal spatial interactions as follows:

$$\mathcal{L}_\delta u = \int_{B_\delta(\mathbf{x})} (u(\mathbf{y}) - u(\mathbf{x})) \gamma_\delta(\mathbf{x}, \mathbf{y}) d\mathbf{y}, \quad (1.1)$$

where δ is a horizon parameter, $B_\delta(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^d : \|\mathbf{y} - \mathbf{x}\| < \delta\}$ is the δ -radius neighborhood of $\mathbf{x}$, the kernel $\gamma_\delta(\mathbf{x}, \mathbf{y}) = \gamma_\delta(\|\mathbf{y} - \mathbf{x}\|)$ is a nonnegative and symmetric radial-type function with a compact support of $\gamma_\delta(s)$ in $[0, \delta]$.

* Corresponding author.

E-mail addresses: haot@ouc.edu.cn (H. Tian), zjing@mail.ccnu.edu.cn (J. Zhang), ju@math.sc.edu (L. Ju).

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In this paper, the following nonlocal diffusion model with volume constrained boundary conditions is the main subject of interest:

$$\begin{cases} -\mathcal{L}_\delta u = f(\mathbf{x}), & \mathbf{x} \in \Omega, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \Omega_s. \end{cases} \quad (1.2)$$

where $\Omega \in \mathbb{R}^d$ is the solution domain and $\Omega_s := \{\mathbf{y} \in \mathbb{R}^d / \Omega : \|\mathbf{y} - \mathbf{x}\| < \delta, \mathbf{x} \in \Omega\}$ is the constrained or interaction domain. The well posedness of (1.2) was thoroughly studied in [6]. Note that under the assumption

$$0 < C = \int_0^\delta s^2 \gamma_\delta(s) ds < \infty, \quad (1.3)$$

when $\delta \rightarrow 0$, the nonlocal operator $\mathcal{L}_\delta$ transforms into the local diffusion operator,

$$\mathcal{L}_\delta u \rightarrow \mathcal{L}_0 u = \Delta u, \quad \forall u \in C^\infty.$$

Within the last few years, a number of numerical methods have been proposed for solving the models associated with the nonlocal operator (1.1), such as the finite difference and finite element methods (cf. [4,9,11,12,15,19,24,27,36,41,43,48]). These schemes are constructed based on local basis functions. Therefore, we refer to them as “local schemes” for the remainder of this paper.

One critical issue is that local schemes have extraordinary computational costs and memory requirements. This issue is partially caused by nonlocal operators. The resulting system matrix of nonlocal operators is not sparse and has a memory cost of $O(N^2)$, where N is the degree of freedom in the discrete system. Conjugate gradient solvers have been commonly applied to solve the discretized system, but such solvers require $O(N^2 \log N)$ operations. Moreover, because of the irregularity of partially intersecting regions (also denoted as partial areas), which lie between the support area of basis functions and the neighborhood of nodes, numerical integrations cannot be computed with high accuracy.

Based on the foregoing observations, it would be desirable to develop a global method to solve nonlocal problems. Spectral methods, which are a class of global methods, have some attractive properties. Firstly, spectral methods can achieve a fixed accuracy with much fewer unknowns. Secondly, unlike local schemes, the basis functions of spectral methods are global, meaning there are no partial areas in these methods. In other words, the numerical integrations of discrete functions on partial areas can be computed exactly. Recently, spectral methods have been broadly applied to solve integral equations, such as the Volterra integral equations [20,38,44,45] and fractional partial differential equations [46,47]. The authors developed a fast Fourier spectral method for nonlocal diffusion equations under periodic boundary conditions in [13,14]. However, spectral methods for nonlocal diffusion models with volume constrained boundary conditions have received very little attention.

In this paper, we provide new insights into spectral collocation algorithms for nonlocal diffusion models with volume constrained boundary conditions. The main contributions of this paper are as follows. We propose spectral collocation methods for solving nonlocal diffusion models on a regular domain and present rigorous error analysis that theoretically justifies the spectral accuracy of convergence and its asymptotic compatibility. This paper is our first step towards efficient high order methods including spectral and spectral element methods for nonlocal diffusion equations with volume constrained boundary conditions.

The remainder of our work is organized as below. In Section 2, we review some fundamental properties of Legendre polynomials and identify the properties to be used throughout this paper. In Section 3, a spectral approach for one-dimensional nonlocal diffusion models is proposed. In Section 4, we present error analysis in L^∞ norms. In Section 5, we demonstrate that the proposed schemes are able to be generalized to high-dimensional cases. Numerical examples are given to justify our theory and demonstrate the efficiency of proposed algorithms in Section 6. In Section 7, we provide several concluding remarks.

2. Notation and preliminaries

In this section, we introduce some notations and review some properties of Legendre polynomials that will be used in the forthcoming sections (cf. [1,37]).

2.1. Notation

- Let $\omega^{\alpha,\beta}(z) = (1-z)^\alpha(1+z)^\beta$, $z \in I := (-1, 1)$, $\alpha, \beta > -1$ be the Jacobi weight function. The inner product of $L^2_{\omega^{\alpha,\beta}}(I)$ is denoted by $(u, v)_{\alpha,\beta} = \int_I u(z)v(z)\omega^{\alpha,\beta}(z)dz$. The norm of u is denoted by $\|u\|_{\omega^{\alpha,\beta}} = \sqrt{(u, u)_{\alpha,\beta}}$. For any integer $r \geq 0$, we introduce the weighted Sobolev space:

$$H^r_{\omega^{\alpha,\beta}}(I) = \left\{ u \in L^2_{\omega^{\alpha,\beta}}(I) : \|u\|_{r,\omega^{\alpha,\beta}} = \left(\sum_{k=0}^r \|u^{(k)}\|_{\omega^{\alpha,\beta}}^2 \right)^{\frac{1}{2}} < \infty \right\}.$$

Therefore, we have $L^2_{\omega^{0,0}}(I) = H^0_{\omega^{0,0}}(I)$. For convenience, its inner product and norm are denoted by $(\cdot, \cdot)$ and $\|\cdot\|$, respectively.

- For $m \in \mathbb{N}$, the non-uniform Jacobi-weighted Sobolev space is defined as

$$B_{\alpha,\beta}^m(I) := \{u : \partial_x^k u \in L_{\omega^{\alpha+k,\beta+k}}^2(I), \quad 0 \leq k \leq m\}.$$

For $m \in \mathbb{N}$, this space has the inner product and norm

$$(u, v)_{B_{\alpha,\beta}^m} = \sum_{k=0}^m (\partial_x^k u, \partial_x^k v)_{\omega^{\alpha+k,\beta+k}}, \quad \|u\|_{B_{\alpha,\beta}^m} = (u, u)_{B_{\alpha,\beta}^m}^{1/2}.$$

- We denote the set of all algebraic polynomials of degree $\leq N$ by $\mathbb{P}_N$.

2.2. Legendre polynomials

The Legendre polynomials, denoted by $\{L_n(z), n \geq 0\}$, are mutually orthogonal with respect to $\omega^{0,0}(z) = 1$, where

$$\int_{-1}^1 L_m(z) L_n(z) dz = \varrho_n \delta_{mn}, \quad \varrho_n = \frac{2}{2n+1},$$

and they can be determined by the three-term recurrence relation:

$$(n+1)L_{n+1}(z) = (2n+1)zL_n(z) - nL_{n-1}(z), \quad n \geq 1, \\ L_0(z) = 1, \quad L_1(z) = z.$$

2.3. Legendre-Gauss-Type quadratures

We now review some Gauss-Type Quadrature formulas:

- For $0 \leq j \leq N$, the Legendre–Gauss–Lobatto quadrature is

$$\{x_j\}_{j=0}^N \text{ are the zeros of } (1-x^2)L'_N(x); \quad \omega_j = \frac{2}{N(N+1)} \frac{1}{[L_N(x_j)]^2}.$$

- For $0 \leq j \leq N$, the Legendre–Gauss quadrature is

$$\{x_j\}_{j=0}^N \text{ are the zeros of } L_{N+1}(x); \quad \omega_j = \frac{2}{(1-x_j^2)[L'_{N+1}(x_j)]^2}.$$

- For $0 \leq j \leq N$, the Legendre–Gauss–Radau quadrature is

$$\{x_j\}_{j=0}^N \text{ are the zeros of } L_N(x) + L_{N+1}(x); \quad \omega_j = \frac{1}{(N+1)^2} \frac{1-x_j}{[L_N(x_j)]^2}.$$

3. Numerical scheme

To illustrate the algorithm, we now present the Legendre collocation scheme for one-dimensional case of (1.2). First, for $x \in \mathbb{R}^1$, the problem (1.2) is transformed into

$$u(x) \int_{-\delta}^{\delta} \gamma_{\delta}(x, x+s) ds - \int_{x-\delta}^{x+\delta} u(y) \gamma_{\delta}(x, y) dy = f(x), \quad x \in I := (-1, 1). \quad (3.1)$$

by changing the variable

$$y = x + s, \quad s \in (-\delta, \delta).$$

Let Legendre–Gauss–Lobatto points be the collocation points $\{x_i\}_{i=0}^N$ that include both endpoints ± 1 . Then, the approximation to (1.2) using a Legendre collocation scheme can be written as

$$\begin{cases} \text{Find } u_N \in \mathbb{P}_N \text{ such that} \\ u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i+s) ds - \int_{\Lambda^i} u_N(y) \gamma_{\delta}(x_i, y) dy \\ = f(x_i) + \int_{\Lambda_c^i} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N, \end{cases} \quad (3.2)$$

where $\Lambda^i := (x_i - \delta, x_i + \delta) \cap (-1, 1)$ and $\Lambda_c^i := (x_i - \delta, x_i + \delta) / \Lambda^i$. We define $\Lambda^i := (a_i, b_i) = (x_i - \delta, x_i + \delta) \cap (-1, 1)$ and perform the following simple linear transformation:

$$y(x_i, t) = \frac{b_i - a_i}{2} t + \frac{a_i + b_i}{2}, \quad t \in (-1, 1).$$

Then, (3.2) becomes

$$u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{b_i - a_i}{2} \int_{-1}^1 u_N(y(x_i, t)) \gamma_{\delta}(x_i, y(x_i, t)) dt = f(x_i) + \int_{\Lambda_t^i} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N. \quad (3.3)$$

To compute the integral term accurately, we use a Legendre-Gauss-type quadrature rules $\{t_j, \omega_j\}_{j=0}^M$. Then, the Legendre collocation scheme for (3.3) becomes:

$$\begin{cases} \text{Find } u_N \in \mathbb{P}_N \text{ such that} \\ u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{b_i - a_i}{2} \sum_{j=0}^M u_N(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j \\ = f(x_i) + \int_{\Lambda_t^i} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N. \end{cases} \quad (3.4)$$

We seek the approximate solution u_N in the form:

$$u_N(x) = \sum_{k=0}^N u_k h_k(x),$$

where $\{h_k\}$ are the Lagrange basis polynomials (i.e., $h_k \in \mathbb{P}_N$ and $h_k(x_j) = \delta_{kj}$). Inserting this solution into (3.4) yields

$$\begin{aligned} u_i \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{b_i - a_i}{2} \sum_{k=0}^N u_k \left(\sum_{j=0}^M h_k(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j \right) \\ = f(x_i) + \int_{\Lambda_t^i} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N. \end{aligned} \quad (3.5)$$

We denote $\mathbf{u}_N = (u_0, u_1, \dots, u_N)^T$. Then, we obtain the matrix form of the Eq. (3.5):

$$\mathbf{A} \mathbf{u}_N = \mathbf{F},$$

where the entries of the matrix $\mathbf{A} = (A_{i,j})_{0 \leq i,j \leq N}$ can be computed by the explicit formulas below

$$A_{i,j} = \begin{cases} -\frac{b_i - a_i}{2} \sum_{j=0}^M h_k(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j, & i \neq j, \\ \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{b_i - a_i}{2} \sum_{j=0}^M h_k(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j, & i = j. \end{cases}$$

and the entries of the vector $\mathbf{F}$ are

$$F_i = f(x_i) + \int_{\Lambda_t^i} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N.$$

More precisely, we divide the integral range of $(x_i - \delta, x_i + \delta)$ into three case.

Case I: $-1 < x_i - \delta < x_i + \delta < 1$. In this case, we convert the interval $(x_i - \delta, x_i + \delta)$ into $(-1, 1)$ by using the following linear transformation:

$$y = x_i + \delta t, \quad t \in (-1, 1).$$

The scheme then becomes

$$u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \delta \int_{-1}^1 u_N(y(x_i, t)) \gamma_{\delta}(x_i, y(x_i, t)) dt = f(x_i), \quad 0 \leq i \leq N. \quad (3.6)$$

Next, we compute the integral term using a Legendre-Gauss-type quadrature $\{t_j, \omega_j\}_{j=0}^M$, yielding

$$u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \delta \sum_{j=0}^M u_N(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j = f(x_i), \quad 0 \leq i \leq N. \quad (3.7)$$

We expand the approximate solution u_N by using the Lagrange basis $\{h_k\}_{k=0}^N$,

$$u_N(x) = \sum_{k=0}^N u_k h_k(x).$$

Inserting this solution into (3.7) yields

$$u_i \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \delta \sum_{k=0}^N u_k \sum_{j=0}^M h_k(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j = f(x_i).$$

Case II: $x_i - \delta < -1 < x_i + \delta < 1$. In this case, we have

$$u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \int_{-1}^{x_i + \delta} u_N(y(x_i, t)) \gamma_{\delta}(x_i, y) dy = f(x_i) + \int_{x_i - \delta}^{-1} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N.$$

Next, we convert the interval $(-1, x_i + \delta)$ into $(-1, 1)$ and use the Legendre-Gauss-type quadrature $\{t_j, \omega_j\}_{j=0}^M$ to approximate the integral term accurately. More precisely, we have

$$\begin{aligned} u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{x_i + \delta + 1}{2} \sum_{j=0}^M u_N(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j \\ = f(x_i) + \int_{x_i - \delta}^{-1} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N. \end{aligned} \quad (3.8)$$

Then, substituting approximate solution $u_N(x) = \sum_{k=0}^N u_k h_k(x)$ into (3.8) yields

$$\begin{aligned} u_i \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \frac{x_i + \delta + 1}{2} \sum_{k=0}^N u_k \sum_{j=0}^M h_k(y(x_i, t_j)) \gamma_{\delta}(x_i, y(x_i, t_j)) \omega_j \\ = f(x_i) + \int_{x_i - \delta}^{-1} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N. \end{aligned}$$

Case III: $-1 < x_i - \delta < 1 < x_i + \delta$. In this case, we have

$$u_N(x_i) \int_{-\delta}^{\delta} \gamma_{\delta}(x_i, x_i + s) ds - \int_{x_i - \delta}^1 u_N(y(x_i, t)) \gamma_{\delta}(x_i, y) dy = f(x_i) + \int_{-1}^{x_i - \delta} g(y) \gamma_{\delta}(x_i, y) dy, \quad 0 \leq i \leq N.$$

Therefore, we can treat this case the same as the case above.

Remark 3.1. In spectral collocation method, we write

$$h_k(x) = \sum_{p=0}^N \beta_{p,k} L_p(x), \quad 0 \leq p, k \leq N,$$

and determine the coefficients $\beta_{p,k}$ from $h_k(x_j) = \delta_{kj}$, $0 \leq k, j \leq N$. More precisely,

$$\beta_{p,k} = \frac{1}{\gamma_p} \sum_{i=0}^N h_k(x_i) L_p(x_i) \omega_i / \varrho_p = L_p(x_k) \omega_k / \varrho_p,$$

where

$$\varrho_p = \sum_{i=0}^N L_p^2(x_i) \omega_i = \left(p + \frac{1}{2}\right)^{-1}, \quad \text{for } p < N,$$

and $\varrho_N = 2/N$ for the Gauss-Lobatto formula, and $\varrho_N = (N + 1/2)^{-1}$ for the Gauss and Gauss-Radau formulas.

Remark 3.2. It is noteworthy that we can choose the different type and number of collocation points $\{x_j\}_{j=0}^N$ and quadrature points $\{t_j\}_{j=0}^M$.

4. Convergence analysis and asymptotic compatibility

In this section, we analyze the convergence of the proposed scheme and demonstrate that asymptotic compatibility of the spectral collocation scheme.

4.1. Some useful lemmas

Lemma 4.1 [28, Lemma 4.8]. If $u \in B_{-1,-1}^m(I)$ with $1 \leq m \leq N + 1$, then for any $\phi \in \mathbb{P}_N$,

$$|(u, \phi) - \langle u, \phi \rangle_N| \leq c \sqrt{\frac{(N - m + 1)!}{N!}} (N + m)^{-(m+1)/2} \|\partial_x^m u\|_{\omega^{m-1, m-1}} \|\phi\|,$$

where c is a positive constant independent of u , m , N and ϕ .

Lemma 4.2 [28, B.33]. Let (a, b) be a finite interval. Then, the following Sobolev inequality holds:

$$\max_{x \in [a, b]} |u(x)| \leq \left(\frac{1}{b-a} + 2\right)^{1/2} \|u\|_{L^2(a, b)}^{1/2} \|u\|_{H^1(a, b)}^{1/2}, \quad \forall u \in H^1(a, b). \quad (4.1)$$

This inequality is also known as the Gagliardo-Nirenberg interpolation inequality.

Lemma 4.3 [28, Corollary B.2]. For any $u \in H^1(a, b)$ with $u(x_0) = 0$, for some $x_0 \in (a, b)$, the following Poincare inequality holds:

$$\|u\|_{L^2(a,b)} \leq c \|u'\|_{L^2(a,b)}, \quad \forall u \in H^1(a, b). \quad (4.2)$$

Lemma 4.4 [28, Theorem 3.44]. For any $u \in B_{-1,-1}^m(I)$ and $1 \leq m \leq N+1$, we have

$$\|\partial_x(I_N u - u)\| + N \|I_N u - u\|_{\omega^{-1,-1}} \leq c \sqrt{\frac{(N-m+1)!}{N!}} (N+m)^{(1-m)/2} \|\partial_x^m u\|_{\omega^{m-1,m-1}},$$

where c is a positive constant independent of m , N , and u .

4.2. Maximum principle

It is known that the local diffusion equation satisfies the maximum principle. In comparison, the proposed nonlocal diffusion operator (1.1) satisfies a similar property, see [7]. For the sake of clarity, we now demonstrate that the proposed nonlocal diffusion operator (1.1) satisfies the maximum principle in one dimensional case. This property play an important part in the error analysis.

Theorem 4.1 (Maximum principle). Suppose that $\mathcal{L}_\delta u$ in (1.1) is well-defined in $I := (-1, 1)$. It holds for $\mathcal{L}_\delta u \geq 0$ in I that a maximum of u is attained in the interaction domain $I_c := (-1 - \delta, -1] \cup [1, 1 + \delta)$.

Proof. Consider an auxiliary function

$$v(x) = u(x) + \epsilon x^2, \quad (4.3)$$

with arbitrary constants $\epsilon > 0$. Because the kernel $\gamma_\delta(|y-x|)$ is a symmetric and nonnegative radial-type function, we derive

$$\mathcal{L}_\delta v(x) = \mathcal{L}_\delta u(x) + \epsilon \int_{-\delta}^{\delta} (2xs + s^2) \gamma_\delta(|s|) ds = \mathcal{L}_\delta u(x) + \epsilon \int_{-\delta}^{\delta} s^2 \gamma_\delta(|s|) ds > 0,$$

where $\mathcal{L}_\delta$ is defined by (1.1).

We claim that $v(x)$ cannot attain a maximum in I . If it can, we assume that a nonnegative maximum of v is attained at $x_0 \in I$ (i.e., $v(x_0) = \max_{x \in I} v(x) \geq 0$). Then, it is easy to verify that

$$\mathcal{L}_\delta v(x_0) = \int_{x_0-\delta}^{x_0+\delta} \overbrace{(v(y) - v(x_0))}^{\leq 0} \overbrace{\gamma_\delta(|y-x_0|)}^{\geq 0} dy \leq 0,$$

which leads to a contradiction. This implies that v attains a nonnegative maximum at $x_0 \in I_c$. Based on (4.3), by setting ϵ to zero, we can derive the required result. $\square$

Lemma 4.5. Assume that

$$0 < C = \int_0^\delta s^2 \gamma_\delta(s) ds < \infty, \quad \delta > 0. \quad (4.4)$$

Let $p(t)$ and $G(t)$ be an integrable function that satisfies

$$p(t) \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds - \int_{t-\delta}^{t+\delta} p(y) \gamma_\delta(|y-t|) dy = G(t), \quad t \in (-1, 1) \quad (4.5)$$

and

$$p(t) = 0, \quad t \in (-1 - \delta, -1] \cup [1, 1 + \delta),$$

Then,

$$\|p(t)\|_{L^\infty(I)} \leq c \|G\|_{L^\infty(I)}.$$

Proof. From (4.5), one can see that

$$|p(t)| \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds - \int_{t-\delta}^{t+\delta} |p(y)| \gamma_\delta(|y-t|) dy \leq \|G(t)\|_{L^\infty(I)}.$$

By denoting $E(\eta) := \frac{|p(\eta)|}{\|G(t)\|_{L^\infty(I)}}$, $\eta \in (-1 - \delta, 1 + \delta)$, we can rewrite above inequality as

$$\int_{t-\delta}^{t+\delta} E(y) \gamma_\delta(|y-t|) dy - E(t) \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds \geq -1. \quad (4.6)$$

For $A(t) := [\frac{t(1-t)}{2} + \delta(1+\delta)]/C$, $t \in (-1-\delta, 1+\delta)$, it holds that

$$\int_{t-\delta}^{t+\delta} A(y) \gamma_\delta(|y-t|) dy - A(t) \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds = -\frac{1}{C} \int_0^\delta s^2 \gamma_\delta(|s|) ds = -1. \quad (4.7)$$

Based on (4.6) and (4.7), we can derive that

$$\int_{t-\delta}^{t+\delta} B(y) \gamma_\delta(|y-t|) dy - B(t) \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds \geq 0, \quad t \in (-1, 1),$$

where $B(y) := E(y) - A(y)$, $y \in (-1-\delta, 1+\delta)$. Since $p(t) = 0$ for $t \in (-1-\delta, -1] \cup [1, 1+\delta)$, we obtain that $E(t) = 0$, $t \in (-1-\delta, -1] \cup [1, 1+\delta)$. Therefore,

$$B(t) := E(t) - A(t) = -A(t) \leq 0, \quad t \in (-1-\delta, -1] \cup [1, 1+\delta).$$

Consequently, according to Theorem 4.1, we have

$$B(t) \leq 0, \quad \forall t \in (-1, 1), \quad \text{i.e., } E(t) \leq \|A(t)\|_{L^\infty(I)}, \quad \forall t \in (-1, 1).$$

Let $c = \|A(t)\|_{L^\infty(I)}$. Then, we have

$$\|p(t)\|_{L^\infty(I)} \leq c \|G\|_{L^\infty(I)}, \quad \forall t \in (-1, 1),$$

where c is a positive constant independent of p and G . $\square$

We are now ready to derive error estimates.

4.3. Error analysis in L_∞

For convenience of analysis, the collocation and quadrature points in (3.4) are chosen as the Legendre–Gauss–Lobatto type with $M = N$. We treat the other cases similarly.

Theorem 4.2. Let $u(x)$ be the exact solution of (1.2) and assume that

$$u_N(x) = \sum_{i=0}^N u_i h_i(x), \quad \forall x \in I,$$

where u_i is given by (3.5) and $h_i(x)$ is the i -th Lagrange basis function associated with the Legendre–Gauss–Lobatto points $\{x_i\}_{i=0}^N$. If

$$u \in B_{-1,-1}^m(I), \quad \gamma_\delta(x, y) \in L^\infty(I; B_{-1,-1}^k(I)) \cap L^\infty(D), \quad \partial_x \gamma_\delta(x, y) \in L^\infty(D),$$

where $D = \{(x, y) : -1 < x, y < 1\}$ and $1 \leq k, m \leq N+1$, then

$$\begin{aligned} \|u - u_N\|_{L^\infty(I)} &\leq c \sqrt{\frac{(N-k+1)!}{N!}} (N+k)^{-k/2} \|\partial_y^k \gamma_\delta(x, \cdot)\|_{\omega^{k-1,k-1}} \|u\| \\ &\quad + c \sqrt{\frac{(N-m+1)!}{N!}} (N+m)^{-m/2} \|\partial_x^m u\|_{\omega^{m-1,m-1}}, \end{aligned} \quad (4.8)$$

provided that N is sufficiently large, where c is a positive constant independent of m, k, N and u .

Proof. Assume that

$$\bar{u}_N = \begin{cases} u_N(x) = \sum_{i=0}^N u_i h_i(x), & \forall x \in I, \\ u(x) = g(x), & \forall x \in I_c. \end{cases}$$

We begin with (3.4) and reformulate it as

$$u_i \int_{-\delta}^{\delta} \gamma_\delta(|s|) ds - \int_{x_i-\delta}^{x_i+\delta} \bar{u}_N(y) \gamma_\delta(|y-x_i|) dy = f(x_i) + J_1(x_i), \quad x_i \in I, \quad (4.9)$$

where

$$\begin{aligned} J_1(x) &= \frac{b-a}{2} \sum_{j=0}^N u_N(y(x, t_j)) \gamma_\delta(x, y(x, t_j)) \omega_j - \int_{x-\delta}^{x+\delta} \bar{u}_N(y) \gamma_\delta(|y-x|) dy + \int_{\Lambda_c} g(y) \gamma_\delta(x, y) dy \\ &= \frac{b-a}{2} \sum_{j=0}^N u_N(y(x, t_j)) \gamma_\delta(x, y(x, t_j)) \omega_j - \int_{\Lambda} u_N(y) \gamma_\delta(x, y) dy. \end{aligned}$$

Here, $\Lambda := (a, b) = (x - \delta, x + \delta) \cap (-1, 1)$ and $\Lambda_c := (x - \delta, x + \delta) / \Lambda$.

We denote by I_N the Legendre–Gauss–Lobatto interpolation operator. Multiplying (4.9) by $h_l(x)$ and summing from 0 to N , we have

$$u_N(x) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds - I_N \left(\int_{x-\delta}^{x+\delta} \bar{u}_N(y) \gamma_{\delta}(|y-x|) dy \right) = I_N(f) + I_N(J_1), \quad x \in I. \quad (4.10)$$

Clearly, based on (3.1),

$$I_N(f) = I_N(u) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds - I_N \left(\int_{x-\delta}^{x+\delta} u(y) \gamma_{\delta}(|y-x|) dy \right). \quad (4.11)$$

Denote

$$\bar{e}(y) = \begin{cases} e(y) := u_N(y) - u(y), & \forall y \in I, \\ 0, & \forall y \in I_c. \end{cases}$$

Then, inserting (4.11) into (4.10) yields the following error equation:

$$\begin{aligned} (u_N - u)(x) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds - I_N \left(\int_{x-\delta}^{x+\delta} \bar{u}_N(y) \gamma_{\delta}(|y-x|) dy \right) \\ = (I_N(u) - u)(x) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds - I_N \left(\int_{x-\delta}^{x+\delta} u(y) \gamma_{\delta}(|y-x|) dy \right) + I_N(J_1), \quad x \in I. \end{aligned}$$

Therefore,

$$e(x) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds - \int_{x-\delta}^{x+\delta} \bar{e}(y) \gamma_{\delta}(|y-x|) dy = I_N(J_1) + J_2(x) + J_3(x),$$

where

$$\begin{aligned} J_2(x) &= (I_N u - u) \int_{-\delta}^{\delta} \gamma_{\delta}(|s|) ds, \\ J_3(x) &= I_N \left(\int_{x-\delta}^{x+\delta} \bar{e}(y) \gamma_{\delta}(|y-x|) dy \right) - \int_{x-\delta}^{x+\delta} \bar{e}(y) \gamma_{\delta}(|y-x|) dy \\ &= I_N \left(\int_{\Lambda} e(y) \gamma_{\delta}(|y-x|) dy \right) - \int_{\Lambda} e(y) \gamma_{\delta}(|y-x|) dy. \end{aligned}$$

It is worthwhile to point out that $\bar{e}(y) = u(y) - u_N(y) = 0$ for $y \in I_c$. Therefore, we restrict our attention below to $y \in \Lambda = (x - \delta, x + \delta) \cap (-1, 1)$.

According to Lemma 4.5,

$$\|e(x)\|_{L^\infty(I)} \leq c(\|I_N J_1\|_{L^\infty(I)} + \|J_2\|_{L^\infty(I)} + \|J_3\|_{L^\infty(I)}). \quad (4.12)$$

Now, we turn to the estimation of the right-hand side of (4.12). By Lemma 4.1,

$$\begin{aligned} |J_1(x)| &= \left| \frac{b-a}{2} \left(\int_{-1}^1 u_N(y(x, \cdot)) \gamma_{\delta}(x, y(x, \cdot)) dt - \sum_{j=0}^N u_N(y(x, t_j)) \gamma_{\delta}(x, y(x, t_j)) \omega_j \right) \right| \\ &\leq c \sqrt{\frac{(N-k+1)!}{N!}} (N+k)^{-\frac{k+1}{2}} \times \frac{b-a}{2} \|\partial_t^k \gamma_{\delta}(x, y(x, \cdot))\|_{\omega^{k-1, k-1}} \|u_N(y(x, \cdot))\|. \end{aligned}$$

Note that $y \in \Lambda := (a, b) = (x - \delta, x + \delta) \cap (-1, 1) \subseteq (-1, 1)$. Therefore, applying the linear transformation

$$y = x + \frac{b-a}{2}t, \quad t \in (-1, 1),$$

and performing direct calculation yields

$$\begin{aligned} \|\partial_t^k \gamma_{\delta}(\cdot)\|_{\omega^{k-1, k-1}}^2 &= \int_{-1}^1 |\partial_t^k \gamma_{\delta}(x, y(x, t))|^2 (1-t^2)^{k-1} dt \\ &= \frac{b-a}{2} \int_{\Lambda} |\partial_y^k \gamma_{\delta}(x, y)|^2 \left(x + \frac{b-a}{2} - y\right)^{k-1} \left(\frac{b-a}{2} + y - x\right)^{k-1} dy \\ &\leq \|\partial_y^k \gamma_{\delta}(x, \cdot)\|_{\omega^{k-1, k-1}}^2, \end{aligned}$$

and

$$\frac{b-a}{2} \|u_N(y(x, \cdot))\|^2 = \int_{\Lambda} |u_N(y)|^2 dy \leq \|u_N\|^2.$$

Therefore, we obtain an estimate for $|J_1|$: as

$$|J_1(x)| \leq c \sqrt{\frac{(N-k+1)!}{N!}} (N+k)^{-(k+1)/2} \|\partial_y^k \gamma_\delta(x, \cdot)\|_{\omega^{k-1, k-1}} \|u_N\|$$

In the analysis below, we use the following asymptotic estimate of the Lebesgue constant (cf, [26]):

$$\max_{|x| \leq 1} \sum_{j=0}^N |h_j(x)| \simeq \sqrt{N}, \quad N \gg 1.$$

This implies that

$$\begin{aligned} \|I_N J_1\|_{L^\infty} &\leq c \|J_1\|_{L^\infty} \max_{|x| \leq 1} \sum_{j=0}^N |h_j(x)| \\ &\leq c \sqrt{\frac{(N-k+1)!}{N!}} (N+k)^{-k/2} \|\partial_y^k \gamma_\delta(x, \cdot)\|_{\omega^{k-1, k-1}} \|u_N\| \\ &\leq c \sqrt{\frac{(N-k+1)!}{N!}} (N+k)^{-k/2} \|\partial_y^k \gamma_\delta(x, \cdot)\|_{\omega^{k-1, k-1}} (\|e\| + \|u\|). \end{aligned} \quad (4.13)$$

By using the inequalities (4.1) and (4.2), from Lemma 4.4, we obtain that

$$\begin{aligned} \|J_2\|_{L^\infty} &\leq c \|u - I_N u\|^{1/2} \|\partial_x(u - I_N u)\|^{1/2} \\ &\leq c \sqrt{\frac{(N-m+1)!}{N!}} (N+m)^{-m/2} \|\partial_x^m u\|_{\omega^{m-1, m-1}}. \end{aligned} \quad (4.14)$$

Furthermore, using Lemma 4.4 with $m = 1$ yields

$$\begin{aligned} \|J_3\| &\leq c N^{-1} \|\int_{\Lambda} e(y) \gamma_\delta(|y-x|) dy\| \\ &\leq c N^{-1} \|\gamma_\delta(x, b) e(b) - \gamma_\delta(x, a) e(a) + \int_{\Lambda} \partial_x \gamma_\delta(x, y) e(y) dy\| \\ &\leq c N^{-1} (\max_D |\gamma_\delta(x, y)| + \max_D \|\partial_x \gamma_\delta(x, y)\|_{L^\infty}) \|e\|. \end{aligned}$$

Then, we have

$$\begin{aligned} \|J_3\|_{L^\infty} &\leq c \|J_3\|^{1/2} \|\partial_x J_3\|^{1/2} \leq c N^{-1/2} \|e\|^{1/2} \|\partial_x \int_{x-\delta}^{x+\delta} \gamma_\delta(x, y) e(y) dy\| \\ &\leq c N^{-1/2} \|e\| \leq c N^{-1/2} \|e\|_{L^\infty}. \end{aligned} \quad (4.15)$$

A combination of (4.13), (4.14), and (4.15) provides a final estimate. $\square$

Remark 4.1. For $k = o(N)$ (resp. $m = o(N)$), we have

$$\sqrt{\frac{(N-k+1)!}{N!}} \simeq N^{(1-k)/2} \left(\text{resp. } \sqrt{\frac{(N-m+1)!}{N!}} \simeq N^{(1-m)/2} \right).$$

Therefore, for fixed regularity index k (resp. m), the order of convergence in (4.8) is $O(N^{\frac{1}{2}-k})$ (resp. $O(N^{\frac{1}{2}-m})$).

4.4. Asymptotic compatibility

We now demonstrate that spectral collocation methods are asymptotically compatible for nonlocal diffusion models and can preserve limiting behaviors at a discrete level. Mathematically, this can be formulated as

$$\|u_N - u_{loc}\|_{L^\infty(I)} \rightarrow 0 \text{ as } \delta \rightarrow 0, \quad N \rightarrow \infty, \quad (4.16)$$

where we use u_{loc} to denote the solution for the classic diffusion model:

$$\begin{cases} -\mathcal{L}_0 u = f(\mathbf{x}), & \mathbf{x} \in \Omega, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \Omega_s. \end{cases}$$

Theorem 4.3. Assume that

$$u \in B_{-1,-1}^m(I), \quad \gamma_\delta(x, y) \in L^\infty(D) \cap L^\infty(I; B_{-1,-1}^k(I)), \quad \partial_x \gamma_\delta(x, y) \in L^\infty(D),$$

where $1 \leq k, m \leq N+1$ and $D = \{(x, y) : -1 < x, y < 1\}$. Then, it holds that

$$\|u_N(x) - u_{loc}(x)\|_{L^\infty(I)} = O(N^{\frac{1}{2}-k}) + O(N^{\frac{1}{2}-m}) + O(\delta^2). \quad (4.17)$$

Proof. Note that $\square$

$$-\int_{x-\delta}^{x+\delta} (u(y) - u(x)) \gamma_\delta(x, y) dy = f(x), \quad x \in I, \quad (4.18)$$

and

$$-Cu''_{loc}(x) = f(x), \quad x \in I.$$

By Taylor expansion, we have

$$u(y) - u(x) = u'(x)(y - x) + \frac{1}{2}u''(x)(y - x)^2 + \frac{1}{3!}u'''(x)(y - x)^3 + O((y - x)^4). \quad (4.19)$$

In view of (1.3), we have

$$0 < \int_{x-\delta}^{x+\delta} |y - x|^2 \gamma_\delta(x, y) dy = 2 \int_0^\delta s^2 \gamma_\delta(s) ds = 2C < \infty,$$

and by substituting (4.19) into (4.18),

$$-\int_{x-\delta}^{x+\delta} (u(y) - u(x)) \gamma_\delta(x, y) dy = -Cu''(x) + O(\delta^2) = -Cu''_{loc},$$

where C is a constant independent of δ and u . Therefore,

$$\begin{cases} -(u(x) - u_{loc}(x))'' = O(\delta^2), & x \in (-1, 1), \\ u(x) - u_{loc}(x) = 0, & x = \pm 1. \end{cases}$$

Based on the regularity of $u(x) - u_{loc}(x)$, we can derive that $\|u - u_{loc}\|_{1,\omega^{0,0}} \leq O(\delta^2)$. Finally, from the triangle inequality and Theorem 4.2, we have

$$\begin{aligned} \|u_N(x) - u_{loc}(x)\|_{L^\infty(I)} &\leq \|u_N - u\|_{L^\infty(I)} + \|u - u_{loc}\|_{L^\infty(I)} \\ &= O(N^{\frac{1}{2}-k}) + O(N^{\frac{1}{2}-m}) + O(\delta^2). \end{aligned}$$

5. Extension to multidimensional problems

In this section, we present that the proposed schemes and analysis results can easily be generalized for multidimensional problems through the typical tensor product approach.

To fix the idea, we consider

$$\begin{cases} \int_{B_\delta(\mathbf{x})} (u(\mathbf{x}) - u(\mathbf{y})) \gamma_\delta(\|\mathbf{y} - \mathbf{x}\|) d\mathbf{y} = f(\mathbf{x}), & \mathbf{x} \in \Omega, \\ u(\mathbf{x}) = g(\mathbf{x}), & \mathbf{x} \in \Omega_s, \end{cases} \quad (5.1)$$

where Ω is a finite domain in $\mathbb{R}^d$, Ω_s is the boundary zone around Ω , and $B_\delta(\mathbf{x})$ represents a neighborhood of $\mathbf{x}$ with a radius of δ :

$$B_\delta(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^d : \|\mathbf{y} - \mathbf{x}\| < \delta\}.$$

Let us denote $X_N = (\mathbb{P}_N)^d$, we shall construct multi-dimensional basis functions by using the tensor product of one-dimensional basis functions. More precisely, let $\mathbf{x}_i = (x_{1,i}^i, \dots, x_{d,i}^i)^t$ and $\{x_{k,l}^i\}_{l=0}^N$ be a set of Legendre-Gauss-Lobatto points for fixed $k = 1, 2, \dots, d$. We seek an approximate solution in the form of $u_N(\mathbf{x}) \in X_N := \text{span}\{h_{i_1}(x_1) \cdots h_{i_d}(x_d) : i_1, \dots, i_d = 0, \dots, N\}$,

$$u_N(\mathbf{x}) = \sum_{0 \leq i_k \leq N, 1 \leq k \leq d} \hat{u}_{i_1, \dots, i_d} h_{i_1}(x_1) \cdots h_{i_d}(x_d),$$

such that

$$u_{i_1, \dots, i_d} \int_{B_\delta(\mathbf{x}_i)} \gamma_\delta(\|\mathbf{y} - \mathbf{x}_i\|) d\mathbf{y} - \int_{\Lambda_i^c} u_N(\mathbf{y}) \gamma_\delta(\|\mathbf{y} - \mathbf{x}_i\|) d\mathbf{y} = f(\mathbf{x}_i) - \int_{\Lambda_i^c} g(\mathbf{x}) \gamma_\delta(\|\mathbf{y} - \mathbf{x}_i\|) d\mathbf{y}, \quad 0 \leq i \leq N,$$

where $\Lambda^i := B_\delta(\mathbf{x}_i) \cap \Omega$ and $\Lambda_c^i := B_\delta(\mathbf{x}_i) / \Lambda^i$. Then, we can treat this case in the same fashion as the one-dimensional case. Higher dimensional problems can be handled in the same manner.

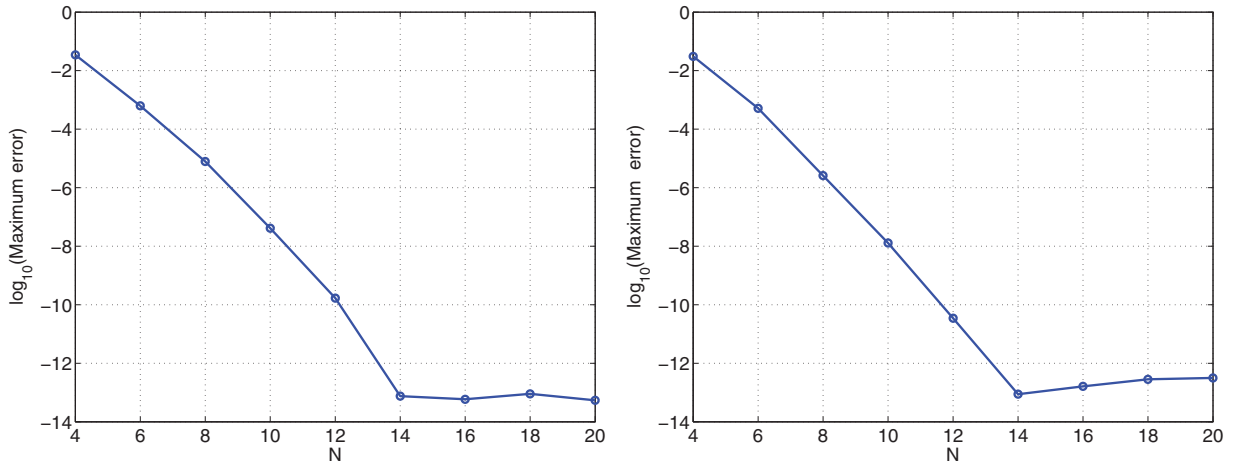


Fig. 6.1. Maximum point-wise errors versus N with $\delta = \frac{1}{16}$ for the Example 6.1. Left: Constant kernel, Right: Gaussian-type kernel.

6. Numerical experiments

In this section, we present some numerical results using the Legendre collocation scheme to verify our theoretical statements. Without loss of generality, we use Legendre–Gauss–Lobatto points as collocation points since the other two types of Legendre–Gauss points can achieve a similar accuracy.

Example 6.1. In the first example, we investigate the convergence properties of numerical solutions. We take $\Omega = (0, 1)$ and consider the one-dimensional nonlocal diffusion Eq. (1.2) with an exact solution $u(x) = xe^x$. We impose nonlocal boundary condition $g(x)$ by using the exact solution. Two different kernel functions are tested below. One is a constant kernel function and the other is a Gaussian-type kernel function.

- For the constant kernel $\gamma_\delta(|y - x|) = 3/\delta^3$, the corresponding source function is

$$f(x) = 3(e^{x-\delta}(\delta + 1) + e^{x+\delta}(\delta - 1))/\delta^3 - 6xe^x/\delta^2 + 3xe^{x-\delta}(e^{2\delta} - 1)/\delta^3.$$

- For the Gaussian-type kernel

$$\gamma_\delta(x, y) = \exp\left(-\frac{|y - x|^2}{\delta^2}\right),$$

the source function is given by

$$\begin{aligned} f(x) = & \frac{\delta^2}{2}(e^{x-\delta-1} - e^{\delta+x-1}) \\ & + \frac{\pi^{1/2}\delta^3}{4}\left(\operatorname{erf}\left(\frac{\delta}{2} + 1\right) - \operatorname{erf}\left(\frac{\delta}{2} - 1\right)\right)e^{\frac{\delta^2}{4}+x} \\ & + \frac{\pi^{1/2}\delta}{2}x\left(\operatorname{erf}\left(\frac{\delta}{2} + 1\right) - \operatorname{erf}\left(\frac{\delta}{2} - 1\right)\right)e^{\frac{\delta^2}{4}+x} - \pi^{1/2}\delta\operatorname{erf}(1)xe^x. \end{aligned}$$

In Fig. 6.1, we plot $\log_{10}(\text{Maximum error})$ versus $N \in [4, 20]$. One can see that the numerical error decreases rapidly and the desired spectral accuracy is obtained. When $N = 14$, the maximum error decreases to 10^{-14} .

In fact, as δ and $1/N$ approach zero, the approximate solutions of the nonlocal problem (1.2) are expected to converge to those of the corresponding classical local problem. To illustrate this convergence behavior of the proposed Legendre collocation method, we plot some numerical results below.

Example 6.2. As a second example, we investigate asymptotically compatible convergence behavior. Let $\Omega = (-1, 1)$. We consider the classic diffusion equation $-u''(x) = f(x)$ with an exact solution $u(x) = xe^x$. For simplicity, we take a constant kernel function $\gamma_\delta(|y - x|) = 3/\delta^3$. Therefore, we have $C = \int_0^\delta s^2 \gamma_\delta(s) ds = 1$. The right-hand functions for the classic diffusion equation and nonlocal diffusion equation are both given by $f(x) = -xe^x - 2e^x$. Additionally, similar to the previous example, the boundary conditions for both classic equations and nonlocal equations are compatible with the exact solution.

In Fig. 6.2, we plot $\log_{10}(\text{maximum error})$ versus $\log_{10}(\delta)$ between the numerical solution for the nonlocal model and the exact solution for the local model. Since N is large enough, the interpolation error is negligible. One can see that the

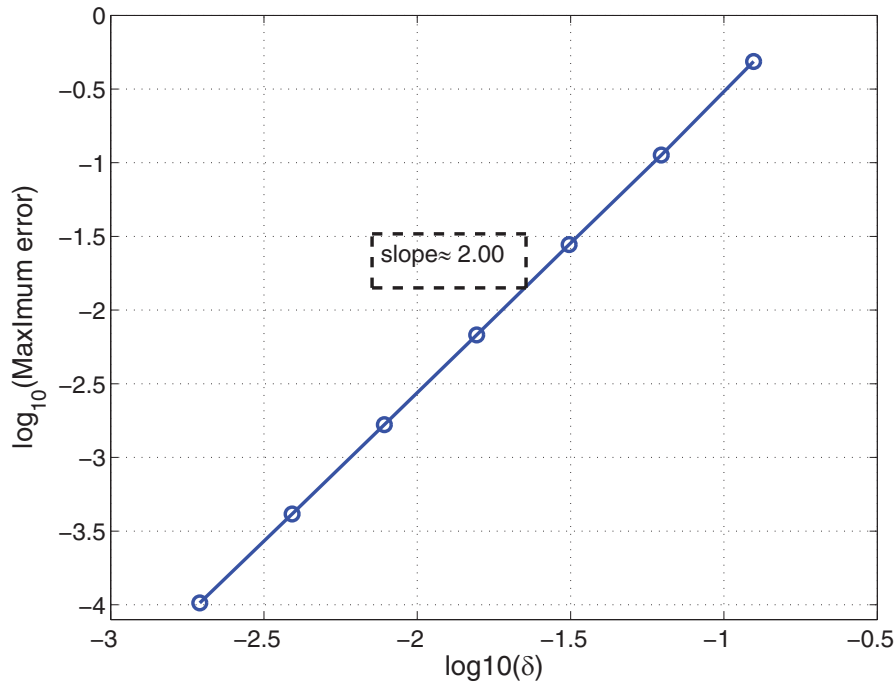


Fig. 6.2. $\log_{10}$ (maximum error) versus $\log_{10}(\delta)$ with $N = 64$.

Table 6.1

Performance of the bilinear collocation method with Gaussian elimination (BC-Gau), the conjugate gradient (BC-CG) method, fast conjugate gradient (BC-FCG) method, and Legendre collocation (LC) method. N is the number of collocation points in one dimension.

N	BC-Gau		N	BC-CG	
	$\ e^h\ _{L_\infty}$	CPU Time		$\ e^h\ _{L_\infty}$	CPU Time
2^4	2.19e-2	0.421s	2^4	2.19e-2	0.078s
2^5	5.72e-3	3m12s	2^5	5.72e-3	2.132s
2^6	1.21e-3	14h41m	2^6	1.21e-3	210.2s
-	-	-	2^7	4.65e-4	10h21m
N	BC-FCG		N	LC	
	$\ e^h\ _{L_\infty}$	CPU Time		$\ e\ $	CPU Time
2^4	2.19e-2	0.051s	4	$2.64e-14$	0.000065s
2^5	5.72e-3	0.187s	-	-	-
2^7	4.65e-4	13.37s	-	-	-
2^8	1.93e-4	2m50s	-	-	-
2^9	9.24e-5	16m30s	-	-	-

slope of the line is approximately 2, as expected. This result verifies Theorem 4.3, which present that the compatibility error is dominated by the third term $O(\delta^2)$ for large values of N .

To illustrate the accuracy and efficiency of the Legendre collocation scheme for nonlocal diffusion model (1.2), we now compare the spectral collocation method and bilinear collocation method for the same model equation (cf. [42]).

Example 6.3. Consider the two-dimensional nonlocal diffusion Eq. (5.1) on the domain $\Omega := (-1, 1)^2$ with a singular kernel

$$\gamma_\delta((x, y), (x', y')) = \frac{1}{\sqrt{(x - x')^2 + (y - y')^2}}.$$

The exact solution is $u(x, y) = x(1 - x)y(1 - y)$. Then, the right-hand side of the nonlocal diffusion problem $f(x, y) = \pi\delta^3(3\delta^2 + 20x^2 + 20y^2 - 20x - 20y)/60$ can be derived according to the terms on the left-hand side. We impose the boundary condition $g(x, y)$ by using the exact solution $u(x, y)$.

In this example, we compare the exact solution with the numerical solution of the nonlocal diffusion Eq. (1.2) discretized with the bilinear collocation method in [42] and the Legendre collocation method. For the bilinear collocation method, we

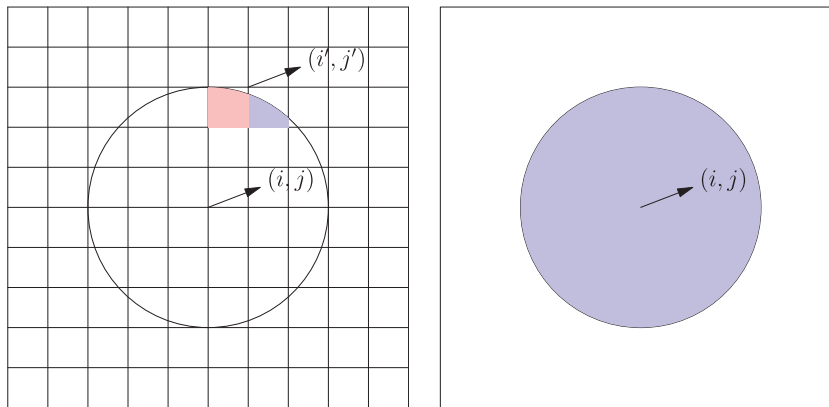


Fig. 6.3. Intersecting areas for different numerical methods. Left: bilinear collocation method, right: spectral collocation method.

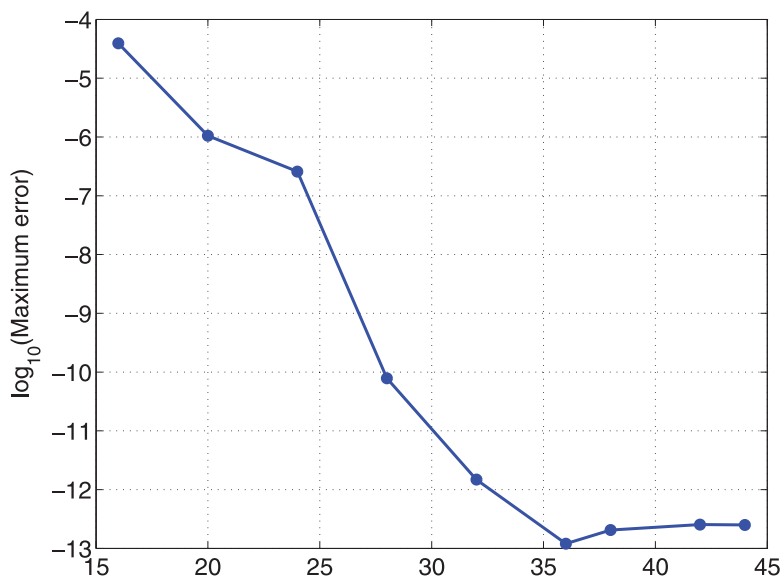


Fig. 6.4. Maximum point-wise errors versus N with $\delta = \frac{1}{16}$ for the Example 6.4.

solve the discrete system using the Gaussian elimination method, conjugate gradient method, and fast conjugate gradient method proposed in [42]. The maximum point-wise errors of the four methods were tabulated in Table 1. One can see that the Legendre collocation method significantly outperforms the bilinear collocation method. The bilinear collocation method requires $2^9 \times 2^9$ collocation points, whereas the Legendre collocation method only requires 5×5 points to decrease the error to $2.64\text{e-}14$. This is typical convergence behavior for a spectral method. The given problem can be resolved spectrally with $N = 4$ based on the regularity of the exact solution.

Remark 6.1. As illustrated in Table 1, the spectral method can achieve a fixed accuracy with much fewer unknowns for the given problem, since its convergence rate is only reduced by the regularity of the underlying function. Another important feature of the Legendre collocation method is that the partial area of the spectral method is a disk or a regular intersecting area between a rectangle and disk in a two-dimensional case. In contrast, the partially intersecting area of the bilinear collocation method is irregular. As shown in Fig. 6.3 (left), in the bilinear collocation method, the blue area and red area represent the intersection between the horizon centered at a point (i, j) and the support area of a bilinear basis function $\phi_{i', j'}$. In Fig. 6.3 (right), for the Legendre collocation method, the support area for the spectral basis function is the entire domain, meaning the intersecting area is the horizon itself. Therefore, numerical integration on the partial area for the Legendre collocation method can be computed exactly. In other words, the serious convergence issue caused by a rough quadrature on the partial area for the bilinear collocation method is no longer a concern for the spectral collocation method.

Example 6.4. Consider the two-dimensional nonlocal diffusion Eq. (5.1) on the domain $\Omega := (-1, 1)^2$ with a singular kernel

$$\gamma_\delta((x, y), (x', y')) = \frac{1}{\sqrt{(x - x')^2 + (y - y')^2}}.$$

The exact solution is $u(x, y) = \exp(-5((x - 1)^2 + (y - 1)^2))$. Then, the right-hand side of the nonlocal diffusion problem is computed by large N . We impose the boundary condition $g(x, y)$ by using the exact solution $u(x, y)$.

As illustrated in Fig. 6.4, we plot $\log_{10}(\text{Maximum error})$ versus $N \in [16, 44]$. One can see that the numerical error decreases rapidly and the desired spectral accuracy is obtained. When $N = 36$, the maximum error decreases to 10^{-13} .

7. Concluding remarks

In this paper, we presented a Legendre collocation scheme for nonlocal diffusion models and provided rigorous error analysis, which theoretically justified the spectral accuracy given that the kernel functions and source functions are sufficiently smooth. It is worthwhile to point out that in spectral collocation methods, the intersecting area between the support region of the basis functions and the neighborhood of the collocation points is regular, meaning accurate numerical integration for assembling stiff mass matrices is not an issue. This makes the spectral collocation methods more attractive than some other classical local methods. As we saw in the error analysis of the Legendre collocation method, the convergence rate is restricted by the regularity of the kernel function. In other words, the Legendre collocation methods exhibit a spectral rate of convergence behavior, the kernel function must be integrable. This fast convergence of spectral method is easily lost if the solution has finite regularity or if the domain is irregular.

However, this work is our first step that we apply a spectral method or a spectral element method to the nonlocal diffusion equation with volume constrained boundary conditions. It is definitely our future work to construct some new spectral methods or spectral element methods which can be applied to problems with less regularities or problems on an irregular domain. We noticed that there have been such efforts in the area of computing (fractional) partial differential equation (cf. [17,29]). Hence, how to solve nonlocal diffusion models with singular kernels on irregular domain by using spectral collocation methods or by spectral element methods is an interesting and important research direction in the future.

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